# ME 5309 – Mid-Semester Examination

Answer Booklet

**Instructions:** Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

## Question 1 [10 marks]

### Q1(a) [4 marks]

(i) f_x = 16/5 = 3.2 pixels, using focal length 16 mm and pixel size 5 microm. (ii) Sensor width is 6.4 mm, so field width = 400x6.4/16 = 160 mm and scale = 0.125 mm/pixel.

### Q1(b) [3 marks]

### Q1(c) [3 marks]

For zero-order hold, F = e^(AT) and G = integral_0^T e^(Aτ)B dτ. Here:

$$A = -2, B = 1, T = 0.1s$$

Thus:

$$F = e^( - 0.2) = 0.818731$$

$$G = (1 - e^( - 0.2))/2 = 0.0906346$$

Therefore:

$$x_{k+1} = 0.818731x_k + 0.0906346u_k$$

With $AT = (-2)(0.1) = -0.2$, $F = e^{-0.2} = 0.8187308$. For $G$, $(1 - F)/2 = (1 - 0.8187308)/2 = 0.1812692/2 = 0.0906346$. Thus the input contribution is positive.

## Question 2 [10 marks]

### Q2(a) [2 marks]

$\dot{x}_1 = x_2$ and $\dot{x}_2 = (\tau - b\, x_2 + mg\ell \sin\theta)/J$. At equilibrium:

$$_0 = -mg\ell sin\theta_e,$$

an equilibrium exists for $|\tau_0|$ less than or equal to 2.4525 N m

### Q2(b) [2 marks]

$$A = [[0, 1], [-mg\ell sin(60°)/J, -b/J]] = [[0, 1], [-106.2, -2.5]],$$

$$B = [[0], [50]]$$

### Q2(c) [4 marks]

At steady state $\dot{\theta} = 0$, and I assume $\theta_{ss} = \theta_d = 90°$ because the PD controller tracks the desired angle. Thus the steady-state error is zero.

### Q2(d) [2 marks]

The gravity term changes the equilibrium, so the steady-state error is still nonzero. The poles should change.

## Question 4 [10 marks]

### Q4(a) [2 marks]

Closed-loop characteristic polynomial is $s^2+(a+bk_2)s+(c+bk_1)=s^2+4s+8$, giving poles -2±2j. At DC, $\theta/r = bN_{ff}/(c+bk_1)$, so unity gain requires $N_{ff}=(c+bk_1)/b=(6+2)/1=8$.

Comparing coefficients with $s^2 + 4s + 8$ gives $a + bk_2 = 4$ and $c + bk_1 = 8$. The feedforward gain is chosen from the nominal plant so the reference scaling is unity; here this gives $N_{ff} = 8$. The closed-loop roots of $s^2 + 4s + 8$ are $-2 \pm 2j$.

### Q4(b) [2 marks]

$$\theta_{ss} = 0.75 \times 8/(6 + 2) = 0.75,$$

the percentage tracking error is 25%

### Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

### Q4(d) [2 marks]